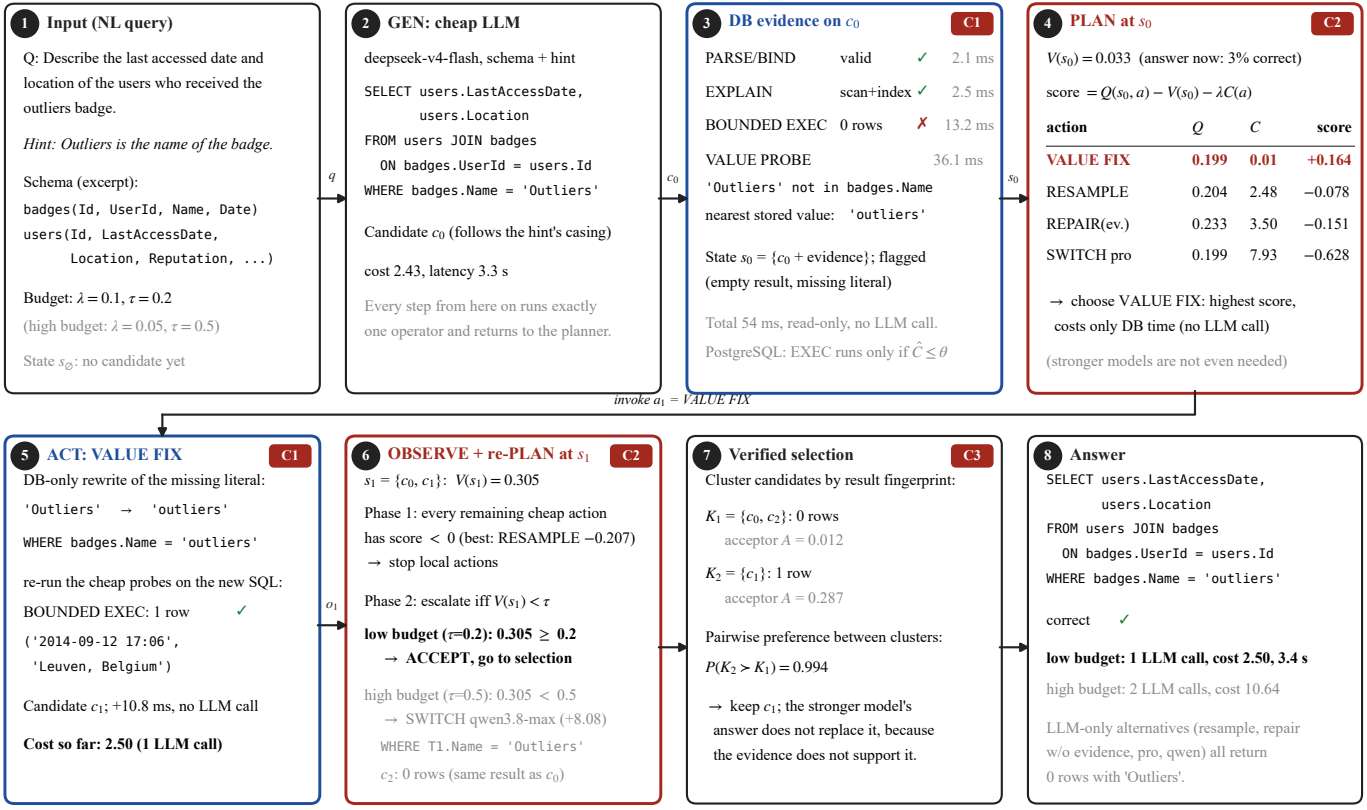

1 Input (NL query)
Q: Describe the last accessed date and location of the users who received the outliers badge.
Hint: Outliers is the name of the badge.
Schema (excerpt):
badges(Id, UserId, Name, Date)
users(Id, LastAccessDate, Location, Reputation, ...)
Budget: λ = 0.1, τ = 0.2
(high budget: λ = 0.05, τ = 0.5)
State s∅: no candidate yet
q
2 GEN: cheap LLM
deepseek-v4-flash, schema + hint
SELECT users.LastAccessDate, users.Location
FROM users JOIN badges
ON badges.UserId = users.Id
WHERE badges.Name = 'Outliers'
Candidate c0 (follows the hint's casing)
cost 2.43, latency 3.3 s
Every step from here on runs exactly one operator and returns to the planner.
c0
3 DB evidence on c0
C1
PARSE/BIND valid ✓ 2.1 ms
EXPLAIN scan+index ✓ 2.5 ms
BOUNDED EXEC 0 rows ✗ 13.2 ms
VALUE PROBE 36.1 ms
'Outliers' not in badges.Name
nearest stored value: 'outliers'
State s0 = {c0 + evidence}; flagged (empty result, missing literal)
Total 54 ms, read-only, no LLM call.
PostgreSQL: EXEC runs only if Ĉ ≤ θ
s0
4 PLAN at s0
C2
V(s0) = 0.033 (answer now: 3% correct)
score = Q(s0, a) − V(s0) − λC(a)
action Q C score
VALUE FIX 0.199 0.01 +0.164
RESAMPLE 0.204 2.48 −0.078
REPAIR(ev.) 0.233 3.50 −0.151
SWITCH pro 0.199 7.93 −0.628
→ choose VALUE FIX: highest score, costs only DB time (no LLM call)
(stronger models are not even needed)
invoke a1 = VALUE FIX
5 ACT: VALUE FIX
C1
DB-only rewrite of the missing literal:
'Outliers' → 'outliers'
WHERE badges.Name = 'outliers'
re-run the cheap probes on the new SQL:
BOUNDED EXEC: 1 row ✓
('2014-09-12 17:06', 'Leuven, Belgium')
Candidate c1; +10.8 ms, no LLM call
Cost so far: 2.50 (1 LLM call)
a1
6 OBSERVE + re-PLAN at s1
C2
s1 = {c0, c1}: V(s1) = 0.305
Phase 1: every remaining cheap action has score < 0 (best: RESAMPLE −0.207)
→ stop local actions
Phase 2: escalate iff V(s1) < τ
low budget (τ=0.2): 0.305 ≥ 0.2
→ ACCEPT, go to selection
high budget (τ=0.5): 0.305 < 0.5
→ SWITCH qwen3.8-max (+8.08)
WHERE T1.Name = 'Outliers'
c2: 0 rows (same result as c0)
7 Verified selection
C3
Cluster candidates by result fingerprint:
K1 = {c0, c2}: 0 rows
acceptor A = 0.012
K2 = {c1}: 1 row
acceptor A = 0.287
Pairwise preference between clusters:
P(K2 ≻ K1) = 0.994
→ keep c1; the stronger model's answer does not replace it, because the evidence does not support it.
8 Answer
SELECT users.LastAccessDate, users.Location
FROM users JOIN badges
ON badges.UserId = users.Id
WHERE badges.Name = 'outliers'
correct ✓
low budget: 1 LLM call, cost 2.50, 3.4 s
high budget: 2 LLM calls, cost 10.64
LLM-only alternatives (resample, repair w/o evidence, pro, qwen) all return 0 rows with 'Outliers'.